# LIB_CLSERV – CLIENT-SERVER LIBRARY FOR PCVUE 16

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| Content: | **LIB_CLSERV** is a library designed to improve the management of PcVue-based Client-Server environments (starting from version 16).<br>It adds functionality for centralized display of the status of remote stations and the distribution of versions, user configuration, and other utilities. |
| Confidentiality: | C0 - Public |

# Content

# Client-Server Library for PcVue 16

## 1. Introduction

**LIB_CLSERV** is a library designed to extend the functionality of PcVue 16 in Client-Server environments, even with redundant architecture.

The library is designed for SCADA architectures in which all PcVue stations share a common Central Folder containing the revisions of projects and libraries through the native Central Project Version Folder functionality, allowing:

- **The status display of all stations**
- **The automatic maintenance of data consistency between server and client;**
- **The centralized distribution of configurations and users;**
- **The reduction of manual maintenance activities and the risks of misalignment between stations.**

## 2. Internal Library Architecture

The logical architecture of LIB_CLSERV is based, in addition to graphic symbols, on a single SCADA-BASIC program called "GENERAL" which contains various subroutines, called in a cyclical and event-based manner. When the project starts, the Main() function:

**Record three programmed cycles (CYCLIC):**

- CheckUpgrade → checks if a reboot of PcVue has been requested on the local station to allow the installation of a new revision of the project and/or libraries;
- Suona → buzzer management (speaker for alarm sound notifications);
- Main2 → secondary initialization of variables after 10 seconds from the start of the station;

**It adds events** to automatically update data such as current user, project version and libraries, connection status based on PcVue's internal variables - which may not be natively distributed if links between stations (e.g. between two clients) have been disabled to optimize and reduce the amount of total sockets opened.

The adoption of the LIB_CLSERV library brings numerous technical and operational advantages:

- **Standardization of Client-Server management logics on all PcVue projects, as it can be easily added to an existing project.**
- **Reduction of maintenance time and manual operations of updating and installing new versions of projects and libraries.**
- **Continuous monitoring of active stations and their operating parameters through** a single mimic that allows you to observe and manage the entire system.
- **Scalability:** Adding new stations only requires the addition of the STATION template instance, with no changes to the underlying logic.
- **Full Traceability:** User login-logoff events on clients are transmitted to PCVUE's centralized logs on servers.
- **Compatibility with RDS and Service environments:** automatic management of PcVue even when running as a service or in remote desktop session with Microsoft Remote Desktop Services (RDS)

# 3. Prerequisites

Before integrating LIB_CLSERV into the PcVue project, make sure that the development environment meets the following technical requirements.

### 1. PcVue Latest Release

We recommend using the latest release of PcVue or in any case higher than PcVue 16.3.1. Make sure that all stations have the same build of PcVue, both on the runtime side and on the editor side, to avoid incompatibilities in the management of global variables and redundant communication services.

### 2. Properly defined redundant architecture

The library is designed for Client-Server environments with real redundancy, in which two server stations manage supervision in parallel (e.g. SRVPCVUE1 and SRVPCVUE2) and multiple clients are connected to both. It is therefore necessary that:

- At least one pair of redundant servers is configured in PcVue - with at least two real-time and historical data associations present (ASSOC_RT and ASSOC_H by default) if there is no redundancy it will be sufficient to remove the symbols relating to the switch between servers from the StationMonitoring synoptic. If there are


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multiple associations, you may want to manually edit the symbols to add the missing variables and commands.

- In case of redundancy, all clients must be configured to access both server stations, so that data is automatically received even in the event of a failover of the main server

## 3. Central Folder

Central Folder is a key element for the library to function. All stations (server and client) must have read and write access to a single shared network folder (SMB), which will serve as a central point for:

- Synchronization of user.dat files (via *user_copy.bat* and *user_load.bat scripts*);
- The distribution of project versions.

Example of a recommended path for the Central Folder: "\\Ip.Address.Server\CentralFolder\"

This avoids DNS issues - provided, of course, that the server containing the shared folder has a static IP. It is perfectly fine to install the shared directory on one of the two servers in order to reduce the perimeter of operation. This folder is not critical as even in the event of unavailability the stations will start, after a short configurable timeout, with the version they have on board.

***Folder configuration requirements:***

- It must be accessible from all stations with the same route.
- NTFS and share permissions must allow read/write to all system users running PcVue (including as a service).

## 4. PcVue Project Structure

The project structure must include the following folders and files to enable library integration:

”C:\ARC Informatique\PcVue 16\Usr\ProjectName”

**|— LIB**

**| └─── LIB_CLSERV\ ← Client-Server Library**

**|— TP\ ← Tools and batches for synchronization**

**| └─── Everything in the folder TP on GitHub**

***Note***:

The TP and LIB_CLSERV folders are an integral part of the library and must be copied to the PcVue project folder to ensure proper operation. Note that starting with version 17 the path of PcVue projects will change as follows:

"C:\ARC Informatique\PcVue 17\data\usr\ProjectName"

# 4. Installation and Configuration

To properly install and configure the LIB_CLSERV library within a PcVue 16 project, follow the steps below precisely. The installation only needs to be done once per project and extended through the creation of a new REF version that will be installed on all stations (server and client).

## 4.1 GitHub to Project integration

**1. Positioning the library**

***Note***: **Make sure PcVue is not running**

With PcVue closed, copy the library folder LIB_CLSERV inside the LIB directory of the PcVue project. Example :

"C:\ARC Informatique\PcVue 16\Usr\ProjectName**\LIB\LIB_CLSERV"**

**2. Creating the User folder in the Central Folder**

Within the network-shared Central Folder, create a new folder named User. This folder will serve as a synchronization point for the user.dat file, which contains the user list and its credentials.

Procedura:

1. Copy the latest version of the user.dat file from your local location:
   C:\ARC Informatique\PcVue 16\Usr\ProjectName**C\user.dat**

2. Paste it into the central path, for example:
   "\\Indirizzo.Ip.Server\CentralFolder\User"

***Tech tip***: Remember to set up sharing of the User folder with read and write permissions for the account used by PcVue.

## 3. Batch File Integration

Copy the TP folder that came with the library to the root directory of the PcVue project. Example: C:\ARC Informatique\PcVue 16\Usr\ProjectName\TP

The TP folder contains auxiliary tools that are essential for the operation of the library, including:

- **user_load.bat** - synchronizes the user.dat file from the Central Folder to the local station;
- **user_copy.bat** - sends the updated user.dat file from the local station to the Central Folder;
- **svrestart.exe** - manages the controlled restart of PcVue (the SCADA-BASIC script launches the svrestart.exe utility by passing it as an argument the PID of the sv32.exe process and the line with which to relaunch PcVue as soon as the current PID closes). It is a file compiled by PcVue - but the sources are available on request;
- **playwav.exe** - manages the playback of alarm WAV files (compiled file)
- **nircmd.exe** - allows the capture of screenshots - comes from https://www.nirsoft.net/utils/nircmd.html

These files must be in the root of the project and not inside the library, as they are called directly from the Basic SCADA code.

## 4. Updating batch scripts

Open the user_copy.bat and user_load.bat files in the TP folder and change the paths according to the project and the location of the Central Folder.

Configuration examples are present within batches.

⚠ ***Caution***:

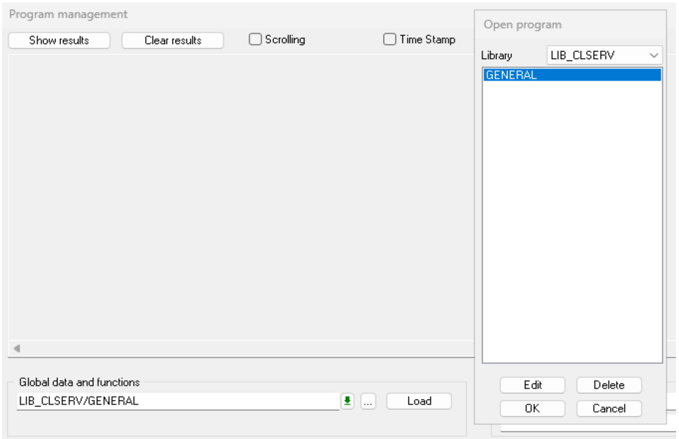
- The routes must be valid and the same on all stations.
- Use full UNC paths (\\Server\Path\...) instead of mapped network drives.
- All stations must have execute and write permissions to the central folder.

***Tip***:
You can test batches manually by double-clicking on them before starting PcVue to verify that the synchronization occurs without access or path errors.

### 5. Verification of the integration

After completing the above steps, **reopen PcVue** and ensure that all library components are properly integrated. Check that:

- The folder LIB_CLSERV is visible in the Application Architect in the Templates section.
- The Basic GENERAL SCADA script is correctly recognized within the LIB (Open Program Management with F9 and open the script to view the various commented functions).




- Batch files in the TP folder can be run manually without errors.
- The User folder in the Central Folder contains a user.dat file that is up-to-date and accessible from the network by all users.

Once everything has been verified, it is possible to proceed with the configuration in Application Architect, where the instances of the stations containing graphic symbols, variables, mimics, etc. will be created.

## 4.2 Configuration by Application Architect

This step is essential to connect templates, create operational instances and ensure that variables and synoptic are correctly referenced in PcVue.

### 1. Creating instances

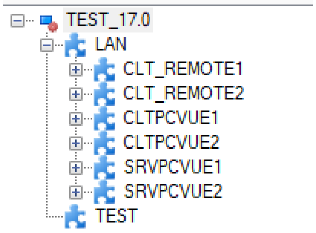
Open Application Architect and create the new instances using the templates provided by the LIB_CLSERV library.

The library provides predefined templates for:

- **The management of network variables (LAN);**
- **Station management;**
- **Alarm Test (TEST).**

For each station in the system (e.g. SRVPCVUE1, SRVPCVUE2, CLT_REMOTE1, CLTPCVUE1, etc.), create an instance that represents the network with the LAN template and then add the stations present through the Station template. Each instance will automatically inherit the variables, mimics, symbols, etc., defined in the template.

***Example:***

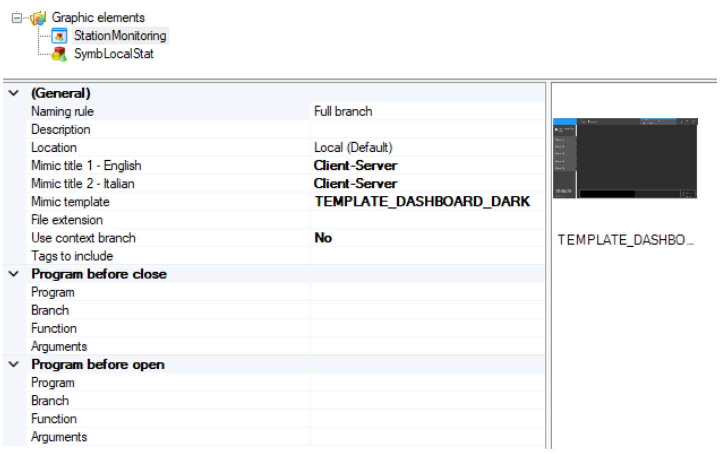



***Technical Note***:
The generated "Station" instances must have names (and Branches) consistent with those defined in the project, in particular in Application Explorer > **Communication > Networking > Stations**

## 2. Adaptation of graphic templates

The LAN template includes a mimic called *StationMonitoring* that uses a Mimic Template called TEMPLATE_DASHBOARD_DARK as the main graphical layout.

If your project uses a Mimic Template, replace the reference in the mimic StationMonitoring with an equivalent template already present in your project.

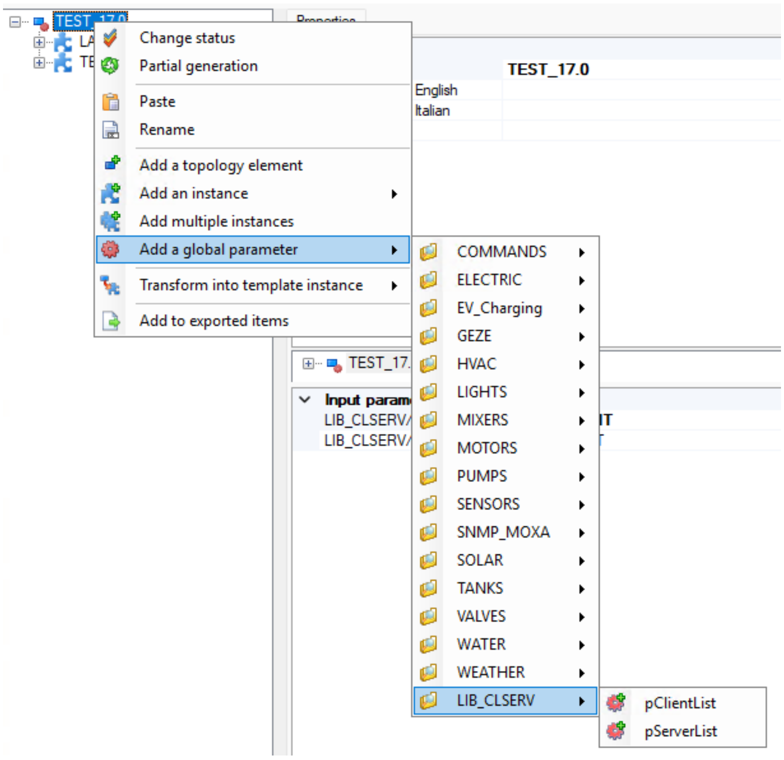


***Function of the mimic StationMonitoring:***

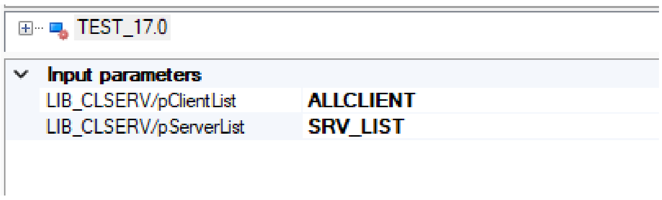
This synoptic acts as a *Control Panel* for the system, showing various information.


## 3. Setting global parameters

In Application Architect, open the main instance (typically named after the project name) and add the required global parameters from the LIB_CLSERV library.




Once the global parameters have been added, it will be necessary to indicate which association they refer to, in the case of example:



**4. Parameter Setting**

In addition to the global parameters, parameters are also used, which must be appropriately corrected on the basis of the associations that the project in question has. So, staying in Application Architect and moving to the **Parameters** section > **LIB_CLSERV** we are going to change the Default Value of the parameters by entering the correct name of the association.

| **(General)** | |
|---|---|
| Name | **pAllServerList** |
| Description 1 - English | |
| Description 2 - Italian | |
| Data type | String |
| Type | Input |
| Default value | **ALLSERVER** |
| Parameter input | Recommended |
| **Validation rule** | |
| Must match property value | No |

**5. Generation and adjustments of the product mimic**

At this point we are ready to always act as Application Architect:

- **File > Save**
- **Task > Generate in Full Synchronization**

Subsequently, we can customize the synoptic produced by Application Architect with the related symbols, in order to have a global view of the SCADA architecture.

**6. Connect the correct popup to the MANAGE USER AND PASSWORD open link**

Here, you will need to Ungroup the symbol shown below and modify the open link with the "PopUp_NetSecurity" mimic found in the LIB_CLSERV library and which allows user and password management.

# 4.3 SCADA Basic Configuration

To make the LIB_CLSERV library operational, you must configure an initialization script in SCADA Basic that enables its functions when the system starts.

**1. Creating the startup script**

Create a new script named "PRELOAD" inside the Local folder and insert the following code:

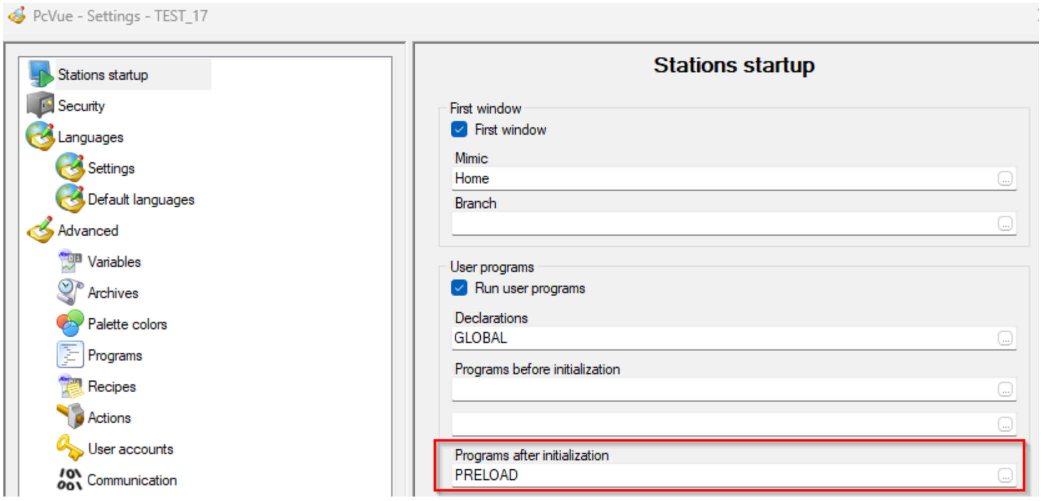
```
Sub Main()

    PROGRAM("PRELOAD", "LIB_CLSERV/GENERAL", "")

    PROGRAM("EXECUTE", "LIB_CLSERV/GENERAL", "")
```

End Sub

In PcVue **Configure** > **Project** > **Settings**, add the PRELOAD function you created in the correct section.




**This script:**

- Automatically starts the GENERAL program contained in the LIB_CLSERV library;
- It ensures that when PcVue starts, the system is ready to handle redundancy, users, and distributed controls.

**2. Customizing the main GENERAL script in the LIB_CLSERV library**

After you create the PRELOAD script, you need **to customize some of the library's Basic SCADA code** to fit your specific project.

***Procedure:***

- **Open the script with F9**
- **Edit redundant bindings:**
  - Update references to real-time and historical associations according to the project configuration.
  - Example: Replace @System.ASSOC_RT. LocalHost with the actual binding defined in the project, if the name differs.
- **Station List**
  - In the BroadcastPassword() routine, update the Stations[] array with the actual names of the stations in the system:

    Stations[1] = "SRVPCVUE1";

    Stations[2] = "SRVPCVUE2";

Stations[3] = "CLT_ENGINEERING";

Stations[4] = "CLT_OPERATOR";

- The names must match those defined in **the Application Architect** (LAN instances and variables).

# 5. Project Launch

After completing all the previous configurations, you can proceed with the **generation and deployment** of the PcVue project.

### 1. Version creation and deployment

In **Configure > Project > Versions**, create a **new version of the project** and distribute it to the other stations via the **shared Central Folder**. Once the deployment is complete, launch PcVue on each station (server and client).
On startup:

- The GENERAL program will be automatically loaded.
- The monitoring CYCLIC **cycles (ControlUpgrade, Ring, Main2)** will be initialized.
- EVENT(s) **related to login/logout, version upgrades, and user logouts** will be logged.
- The LAN.<Station>.* **variables** will be populated for all connected stations.

### Expected result

**Station status**

View the operational status of each server and client in real time:

- **Active**, **Standby**, **Disconnected**
  The LAN.<Station>. ACTIVE and LAN.<Station>. RUNNINGMODE dynamically determine the colors and icons shown in the mimic.

**Connected users**

Shows the list of users currently logged in on each station (LAN.<Station>. USER).

**Operating mode**

Indicates the execution context of each station:

- **Console**

- **Service**
- **RDS (Remote Desktop Session)**
  The value is calculated by the UpgradeControl() program and updated in the RUNNINGMODE field.

**Project Version and Library**

Shows the active version of the **project** (VERSION_P) and **library** (VERSION_L), allowing you to quickly spot any mismatches between nodes after an upgrade or deployment.

**Alarm and sound management**

It allows you to:

- Enable or disable the buzzer via EN_BUZZER and BUZZER,
- Activate or silence alarms from a single station or in centralized mode,
- Manage commands through the Play(), SetPlay(), and ResetPlay() functions.

**User connection control**

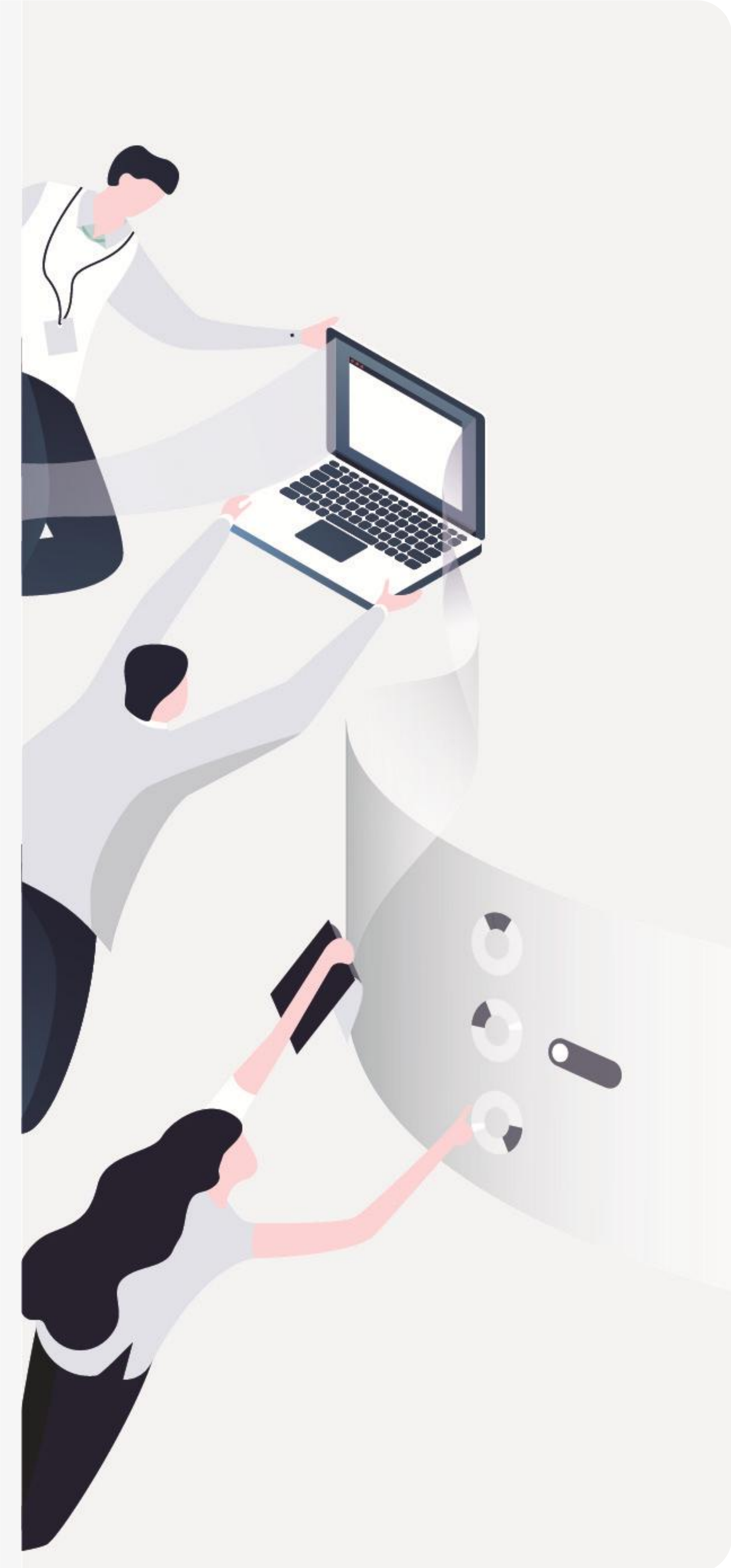
Include command to:

- **Logout remoto** (UserDisconnection()),
- Managing locked or inactive sessions,
- Reset connection triggers (ResetTrigger()).

***Operational Advantage:***
Adding a new station only requires creating a new instance in Application Architect and attaching the symbol in the mimic.

# 6. Conclusion

The LIB_CLSERV library is a strategic extension of the PcVue 16 environment, designed to ensure reliability, consistency, and scalability in redundant SCADA systems. Every station, user and version of the project is tracked, synchronized and supervised in real time, providing a clear view of the status of the system and the interventions carried out.



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